

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – 7

Roll No.: T050	Name: Aashu Yadav
Class: TYIT	Batch: 1
Date of Assignment: 04/08/2026	Date/Time of Submission: 04/08/2026

Aim:-a. Develop a PWA that works in offline mode using Service Worker.

Code:-

manifest.json

```
{
  "name": "HTML TO PWA",
  "short_name": "HTP",
  "start_url": "/",
  "display": "standalone",
  "icons": [
    {
      "src": "icons/icon1.png",
      "sizes": "128x128",
      "type": "image/png"
    },
    {
      "src": "icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png"
    }
  ]
}
```

_Layout.cshtml

```
<link rel="manifest" href="/manifest.json"/>
```

Before head closed

```
<script>
```

```
    if ('serviceWorker' in navigator) {
```

Before body closed

offline.html

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

service-worker.js

```
const CACHE_NAME = "offline-cache-v1";
const urlsToCache = [
  "/",
  "/offline.html",
  "/manifest.json"
];

self.addEventListener("install", event => {
  event.waitUntil(
    caches.open(CACHE_NAME)
      .then(cache => cache.addAll(urlsToCache))
  );
});

self.addEventListener("fetch", event => {
  event.respondWith(
    fetch(event.request)
      .catch(() => caches.match(event.request))
      .then(response => {
        return response || caches.match("/offline.html");
      })
  );
});
```

```
navigator.serviceWorker.register('/service-worker.js')
  .then(() => console.log("Service Worker Registered"))
  .catch(err => console.log("Registration Failed", err)); }
</script>
```

```
<meta charset="UTF-8">
```

```
<title>Offline</title>
```

```
</head>
```

```
<body style="font-family:Arial; text-align:center; margin-top:100px;">
```

```
<h1>You are Offline</h1>
</body>
</html>
```

Output:-

You are Offline

b. Create a Student Record Management PWA.

Code:-

Student.cs (Model):-

```
namespace WebApplication3.Models
{
    public class Student
    {
        public int RollNo { get; set; }
        public string Name { get; set; }
    }
}
```

HomeController.cs :-

```
using Microsoft.AspNetCore.Mvc;
using WebApplication3.Models;
namespace WebApplication8.Controllers
{
    public class HomeController : Controller
    {
        public IActionResult Index()
        {
            Student s = new Student();
            s.RollNo = 50;
            s.Name = "Aashu";
            return View(s);
        }
    }
}
```

Index.cshtml :-

```
@{
    ViewData["Title"] = "Home Page";
}
<div class="text-center">
    <h1 class="display-4">Welcome</h1>
    <p>Learn about <a
href="https://learn.microsoft.com/aspnet/core">building
Web apps with ASP.NET Core</a>.</p>
    <h2>Student Record</h2>
    <p><strong>Roll No :</strong> @Model.RollNo</p>
    <p><strong>Name :</strong> @Model.Name</p>
</div>
```

service-worker.js:-

```
const CACHE = "v1";
self.addEventListener("install", e => {
    e.waitUntil(
        caches.open(CACHE).then(cache => {
            return cache.addAll([
                "/",
```

```
                "/Home/Index",
                "/css/site.css",
                "/js/site.js"
            ]);
        })
    );
});
self.addEventListener("fetch", e => {
    e.respondWith(
        caches.match(e.request).then(response => {
            return response || fetch(e.request);
        })
    );
});
```

manifest.json:-

```
{
    "name": "Student Record PWA",
    "short_name": "StudentPWA",
    "start_url": "/",
    "display": "standalone",
    "background_color": "#ffffff",
    "theme_color": "#0d6efd",
    "icons": [
        {
            "src": "icons/icon1.png",
            "sizes": "128x128",
            "type": "image/png"
        },
        {
            "src": "icons/icon2.png",
            "sizes": "512x512",
            "type": "image/png"
        }
    ]
}
```

_Layout.cshtml:-

```
<link rel="manifest" href="/manifest.json">
(before head closed)
<script>
if ('serviceWorker' in navigator) {
    navigator.serviceWorker.register('/service-worker.js')
}
(Before body closed)
```

output:-

Welcome

Learn about [building Web apps with ASP.NET Core](https://learn.microsoft.com/aspnet/core).

Student Record

Roll No : 50

Name : Aashu